

DL3004 Final Project

Phase 1 - Business Analysis & Dimensional Design

Phase Overview

Duration: Week 1-3

Objective: Design a complete dimensional model before implementation.

Students must understand that a warehouse is designed around business processes and analytical questions, not around copying operational tables.

Aligned Labs

- Lab 3: Logical vs Physical Modeling
- Lab 4: OLAP & Star Schema
- Lab 5: Dimensional Modeling I

Example Business Domain

Domain: E-Commerce Analytics

Business Problem: An online store wants to analyze sales revenue, customer behavior, product performance and regional growth.

Business Processes:

1. Sales Transactions
2. Inventory Snapshots
3. Promotion Events

Example Analytical Questions

Minimum 10 analytical questions.

Sales:

- What are monthly sales trends?
- Which products generate highest revenue?
- Which regions perform poorly?

Customers:

- Which customers generate highest lifetime value?
- Which segments purchase most frequently?

Inventory:

- Which products have declining demand?

3. Identify Business Processes

Students identify warehouse processes.

Example:

Sales Transaction → Fact

Inventory → Periodic Snapshot

Promotions → Factless Fact

4. Define Fact Table Grain

Example:

Fact Sales:

"One row represents one product purchased in one order transaction."

5. Identify Dimensions

Example Star Schema:

Dim Customer

|

Dim Date ---- Fact Sales ---- Dim Product

|

Dim Region

|

Dim Channel

6. Design Star/Snowflake Schema

Students create:

- Conceptual model
- Logical model
- Physical model

Using:

- Draw.io
- dbdiagram.io

Must include:

- Primary keys
- Foreign keys
- Attributes
- Fact measures
- Grain statement

Required Modeling Tasks

- Identify business processes.
- Define grain statements.
- Choose fact table types.
- Identify dimensions and hierarchies.
- Select natural and surrogate keys.
- Create Star or Snowflake schema.

Phase 1 Deliverables

- Project Proposal PDF
- Business Requirements Document PDF
- 10 Business Questions PDF
- Fact & Dimension Identification PDF
- Grain Definition Document PDF
- Star/Snowflake Schema Diagram PNG/PDF
- Data Dictionary Excel

Phase 1 Marks: 30%

Business Understanding: 5

Requirement Analysis: 5

Fact/Dimension Identification: 10

Schema Design: 10

Total: 30